
RAG Chunk-Size Routing: Beyond Fixed Chunks

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Abstract

Retrieval-Augmented Generation (RAG) systems fix corpus chunk size at indexing time, leaving per-query retrieval headroom unrecovered. This project evaluates whether cheap query-only routing can select among candidate chunk sizes and recover a meaningful fraction of the per-question oracle gap without paying the retrieval cost of multi-scale fusion. On a 1.4 MB medical corpus with 398 validated QA pairs spanning factoid, multihop, and synthesis questions, fixed-size retrieval, Reciprocal Rank Fusion (RRF), a per-question oracle, a learned router, and a type-aware heuristic are compared over the action space $\{128, 256, 512\}$. The oracle improves over the best fixed baseline by +8.19 F1 points, confirming that chunk size matters in principle. RRF recovers 12.8% of this gap at higher retrieval cost. However, the learned MiniLM-logistic-regression router underperforms the best fixed baseline, achieving −51.0% gap closure, while the parameter-free type-aware heuristic recovers approximately 20% of the gap. The results suggest that, in this corpus, query-only routing signal is weak and largely captured by question type, while the heavy dominance of 128-token oracle labels structurally limits learned routing performance.

1 Introduction

Retrieval-Augmented Generation (RAG) systems answer queries by grounding a language-model generator on passages retrieved from an external corpus (Lewis et al. 2020, pp. 1-2), conditioning generation on retrieved evidence rather than parametric memory alone to mitigate hallucination (Gao et al. 2024, pp. 1–2). A core but understudied design choice in this pipeline is the *chunk size*: the token budget at which the corpus is segmented before indexing, which sets the granularity of the units a retriever can return (Gao et al. 2024, pp. 7–8). The dominant convention is fixed-size token chunks, inherited from dense-retrieval practice (Karpukhin et al. 2020, ch. 1). Larger chunks preserve more intra-passages context but dilute the retrieval signal and spend more generator tokens per query; smaller chunks concentrate relevant tokens and favour extractive questions, but may strand the answer across chunk boundaries (Gao et al. 2024, p. 8).

Current practice sits at two extremes: fixing a single chunk size by grid search on a held-out set (Gao et al. 2024, pp. 8-10), which accepts average-case suboptimality, or multi-scale retrieval that queries several granularities in parallel and merges them with Reciprocal Rank Fusion (Wang, Ma, et al. 2024, p. 138), improving end-to-end QA quality at higher retrieval cost.

A third strategy lies between the two: a classifier that predicts the chunk size from query-only features. In principle, such a router would deliver single-index retrieval cost while approaching the accuracy of fusion. This strategy has received comparatively little systematic evaluation, and is the focus of this work. To assess whether per-query routing is worth the engineering effort, we cast it as a multiclass classification problem over a small set of candidate chunk sizes and compare the router’s end-to-end RAG F1 to both single-size baselines and RRF fusion. We anchor the comparison to a per-query oracle: a non-learning ceiling that always selects the chunk size which maximises F1 for the question at hand, with deterministic tie-breaking. The oracle is unreachable in deployment — it requires running the full pipeline at every scale and consulting the gold answer — but it bounds what *any* routing policy on this action space can achieve. The *oracle gap* is the F1 difference between the oracle and the best single fixed-size baseline; it quantifies the routable headroom and is the natural unit in which to report adaptive-retrieval gains.

2 Background and Related Work

Open-domain question answering classically separates retrieval from answering (Karpukhin et al. 2020, p. 6769). Sparse lexical scoring such as BM25 remains a reference point (Robertson and Zaragoza 2009, p. 333) and is framed as high-dimensional sparse-vector retrieval (Karpukhin et al. 2020, p. 6770), while Dense Passage Retrieval showed that learned query and passage embeddings could replace hand-engineered lexical matching for evidence selection (Karpukhin et al. 2020, pp. 6769–6771). Our project follows this retrieve-then-answer line: the generator is not trained end to end with the retriever, and the main experimental variable is the evidence passed into it.

Retrieval-augmented generation formalized combining a retriever with a sequence generator for knowledge-intensive tasks (Lewis et al. 2020, pp. 9459–9461). Generators that consume multiple retrieved passages, such as Fusion-in-Decoder, can combine evidence across passages instead of relying on a single context (Izacard and Grave 2021, pp. 874–876). This explains the baseline structure: fixed-size retrieval tests one index at a time, while multi-scale retrieval tests whether evidence from several granularities beats choosing one.

Rank fusion offers a simple non-learned way to combine retrieval outputs: Reciprocal Rank Fusion rewards documents appearing near the top of one or more lists without a trained ranker (Cormack et al. 2009, p. 758). Retrieving from multiple chunk-size indices and merging the ranked evidence is thus a more expensive alternative to selecting one index, and the learned router is positioned against this idea that querying several granularities may be safer than predicting one.

The routing component resembles mixture-of-experts gating, selecting among indices built at different chunk sizes using inexpensive pre-retrieval query features rather than routing to neural expert modules (Jacobs et al. 1991, pp. 79–81). The feature families follow standard text-classification practice: sparse TF-IDF cues, Sentence-BERT embeddings (Reimers and Gurevych 2019, p. 3982), and a compact MiniLM backbone (Wang, Wei, et al. 2020, p. 5776), alongside handcrafted query-only signals. The retriever and generator are held fixed throughout.

3 Method

3.1 Retrieval Pipeline

Every system in this paper shares a single retrieval pipeline; chunk size is the only variable controlled at the system level. Given the raw corpus, we build one chunked corpus and one dense index per candidate size $s \in \mathcal{A} = \{128, 256, 512\}$.

Chunks are produced by a tokenizer-aware sliding window. The corpus is tokenised with the same BGE-small encoder used downstream (Xiao et al. 2024) and emitted in windows of s tokens with stride $s - o$, where the per-size overlap $o = s/8$ guards against entity and sentence boundaries falling exactly at a chunk seam. Each chunk is embedded with BGE-small-en-v1.5, ℓ_2 -normalised (so inner product equals cosine similarity), and inserted into a flat inner-product FAISS index (IndexFlatIP) (Johnson et al. 2017). Flat indexing performs exact search and eliminates approximation noise, isolating chunk size as the only retrieval-side variable; at our corpus scale the additional cost is negligible.

At query time, a system selects a chunk size — fixed for the baselines, predicted by the router, or the full set for fusion — and the query is embedded with the same encoder. To equalise context budgets across sizes we use a size-dependent top- k of $\{20, 10, 5\}$ for $s \in \{128, 256, 512\}$, so every single-scale system delivers approximately $k \cdot s \approx 2,560$ tokens of context to the generator. RRF fusion retrieves $k = 5$ candidates from each per-size index and combines them by

$$s(d) = \sum_{i \in \mathcal{A}} \frac{1}{c + \text{rank}_i(d)}, \quad c = 60,$$

keeping the top-5 fused candidates (Cormack et al. 2009). Retrieved passages are concatenated into a single prompt and answered by Qwen2.5-7B-Instruct (Qwen et al. 2025) with greedy decoding (temperature 0) and a 256-token cap, which makes generation deterministic. The same prompt template and decoding settings are used across every system; only the chunk-size argument differs.

3.2 Oracle Construction

The per-question oracle is a non-learning ceiling computed offline. For each question q and each size $a \in \mathcal{A} = \{128, 256, 512\}$ we run the full retrieve $\rightarrow$ generate $\rightarrow$ score pipeline of §3.1 and record $(F_1(q, a), \text{EM}(q, a))$ on a shared eval grid.

The oracle’s prediction for question q is

$$a^*(q) = \arg \max_{a \in \mathcal{A}} (F_1(q, a), \text{EM}(q, a), -a),$$

under lexicographic ordering: ties on F_1 break to higher EM, then to smaller a (lower cost). This is deterministic under greedy, temperature-zero decoding, with split mean $\bar{F}_1^* = \frac{1}{|Q|} \sum_q F_1(q, a^*(q))$.

The *oracle gap* for $\mathcal{A}$ is

$$\Delta_{\text{oracle}}(\mathcal{A}) = \bar{F}_1^* - \max_{a \in \mathcal{A}} \frac{1}{|Q|} \sum_q F_1(q, a).$$

Because $\bar{F}_1^*$ averages per-question best F1 while the second term is the best mean F1 under one fixed size, a strictly positive Δ_{oracle} means the per-question and population-level argmax disagree on a non-empty subset of questions — the structural condition under which any router can recover headroom.

The oracle consults the gold answer at every a , so it is unreachable in deployment and serves only as the upper bound any policy on $\mathcal{A}$ must respect. Rows for sizes outside $\mathcal{A}$ (e.g. 1024, §4.2) stay in the grid for reproducibility but are excluded from oracle labels, so no system is measured against a size it cannot select.

3.3 Router Formulation

Adaptive chunk selection is modeled as query-only multiclass classification. For each question q_i , the oracle procedure in Section 3.2 assigns a label $y_i = a^*(q_i) \in A$, where $A = \{128, 256, 512\}$. The

router predicts one chunk size before retrieval using only question text and question-level metadata; retrieved chunks, retrieval scores, generated answers, and gold answers are unavailable at test time.

Lightweight routers are evaluated over TF-IDF unigram/bigram features, frozen MiniLM embeddings, and handcrafted features: query length, question-word indicators, a capitalized-span entity proxy, and question type. Classifiers include logistic regression, linear SVM, and LightGBM. Models are trained on oracle labels using natural class priors. Candidate models are first ranked by validation macro-F1 on oracle-label prediction, then the top candidates are re-ranked by downstream validation RAG F1. A threshold sweep optionally falls back to 128 when non-128 confidence is low (analysed in Appendix D). A separate parameter-free type-aware heuristic assigns each question type to its best mean-F1 chunk size. The 1024-token setting remains only for reproducibility and ablations, not the main router action space.

3.4 Evaluation Metrics

Systems are evaluated at two levels: downstream QA quality and router-label prediction. Downstream quality uses exact match (EM) and token-level F1 against the gold answer, averaged over questions; F1 is the primary metric because many valid answers are short spans or paraphrases for which exact match is too strict. Faithfulness is an auxiliary diagnostic of grounding in retrieved context, but all headline comparisons use F1.

To compare adaptive systems against fixed-size retrieval, oracle gap closure is reported. Let $\bar{F}_1(\pi)$ be the mean test F1 of a system or routing policy π , $\bar{F}_1^*$ the oracle mean F1 from Section 3.2, and $\bar{F}_1^{\text{fixed}} = \max_{a \in A} \bar{F}_1(a)$ the best fixed-size baseline. Gap closure is

$$\text{Closure}(\pi) = \frac{\bar{F}_1(\pi) - \bar{F}_1^{\text{fixed}}}{\bar{F}_1^* - \bar{F}_1^{\text{fixed}}}.$$

A value of 1 matches the oracle, 0 matches the best fixed-size baseline, and negative values indicate that routing harms performance relative to the best fixed size.

Retrieval cost is the mean number of retrieved context tokens per question, and gap closure is read together with it rather than as a standalone number. Router macro-F1 on oracle labels is an auxiliary classifier metric only; the objective is downstream RAG F1 and oracle gap closure.

4 Experiments

Experiments use a validated medical QA benchmark derived from the project corpus. After validation and duplicate removal, the dataset contains 398 question-answer pairs across factoid, multihop, and synthesis questions. Splits are stratified by question type with seed 42, yielding 237 training, 77 validation, and 84 test examples; the test set contains 28 questions per type (construction and composition in Appendix E).

Systems differ only in how the chunk size is chosen: a fixed global size, RRF across all sizes, the per-question oracle, the learned router, or the type-aware heuristic. All systems otherwise share the retrieve-generate-score pipeline of §3.1, the same split and evaluation grid, and the same scoring scripts (full configuration in Appendix G).

4.1 Results

The headline comparison across fixed-size retrieval, RRF fusion, the learned router, and the per-question oracle is collected in Appendix A (Table 1 and Figure 1). Among fixed-size systems, 128-token retrieval is the strongest baseline, reaching 0.2128 F1. The oracle reaches 0.2947 F1, producing an oracle gap of +8.19 F1 points and confirming that per-question chunk-size choice has measurable value in principle. RRF fusion recovers 12.8% of this gap, but requires querying multiple chunk-size indices.

The learned router does not recover the oracle gap. Its test F1 is 0.1710, corresponding to a gap closure of −51.0%, meaning that its routing decisions perform worse than always selecting the best fixed chunk size. This indicates that, under the current corpus and QA distribution, cheap query-only features are insufficient to reliably identify the oracle-best chunk size. The type-aware heuristic

reported in Table 2 performs better than the trained router, suggesting that much of the recoverable query-only signal is captured by question type alone.

4.2 Ablations

Action space. The router’s canonical action space is $\{128, 256, 512\}$. Size 1024 is kept in the evaluation grid as an ablation/control, but it is removed from the supervised router target space. This is a deliberate target-design choice: the oracle gap shows that adaptive routing is worth studying, but the router should not be asked to learn a very rare class that is also a weak fixed action. Table 3 quantifies the cost of this simplification. In the full four-size grid, the oracle–baseline gap is +10.61 F1 points. Restricting the router to three sizes reduces the oracle gap to +8.19 points, a sacrifice of 2.42 points overall. This loss is concentrated in multihop and synthesis questions, whereas the factoid gap is nearly unchanged (Figure 6).

Excluding 1024 is justified on economic and statistical grounds: it is the weakest and most expensive single size and the oracle winner for only 6.0% of test questions, so the three-size router is the main system (Appendix C).

Feature families. Across three representations (sparse TF-IDF, frozen MiniLM embeddings, and handcrafted query/type features), MiniLM plus logistic regression is selected: it reaches validation macro-F1 0.416 and RAG F1 0.278 but drops to test macro-F1 0.292 and RAG F1 0.171, so semantic embeddings aided validation selection without generalising past fixed-128, whereas a type-only heuristic reaches F1 0.229. The routing signal is present but not reliably captured by the trained classifier.

Classifier choice. Logistic regression is selected over linear SVM and LightGBM, consistent with the data regime: only 237 training examples and oracle labels heavily skewed toward size 128 give higher-capacity boundaries little minority-class evidence to exploit. Probability-calibration details are deferred to Appendix F.

Per-type routing signal. The per-type ablation locates the routable headroom. Factoid questions hold most of it: their best fixed-size F1 is 0.4863 and their oracle F1 is 0.6326, a gap of +14.63 points (Figure 4); multihop and synthesis gaps are far smaller (+3.02 and +2.05 points). Per-type means for the best fixed baseline, the type-aware heuristic, and the router are near-identical (Figure 5), so the router adds little once question type is fixed.

5 Discussion and Limitations

These results separate adaptive chunking’s potential value from the difficulty of learning it: the positive oracle gap shows per-question chunk choice can help, yet the learned router falls below the strongest fixed baseline. This limits lightweight query-only routing under this corpus and label distribution, but is not evidence against adaptive retrieval generally.

The main constraint is oracle-label imbalance: 128-token chunks are oracle-best for most test questions (Figure 3), so fixed 128 is hard to beat and the usable headroom is small; predicting weakly separated minority sizes can hurt F1, while defaulting to 128 removes adaptivity. The type-aware heuristic again indicates the recoverable signal is mostly question type, with richer embeddings adding little. The QA set compounds this: many questions — especially factoids — are answerable from short local spans, so a more diagnostic benchmark would need distributed-context and comparison questions that fail under small chunks.

6 Conclusion

Chunk size does matter — the per-question oracle beats the best fixed baseline by 8.19 F1 points — but the learned MiniLM + logistic-regression router fell below that baseline, and a type-only heuristic outperformed it, so the useful query-only signal is largely coarse question metadata. Robust routing therefore likely needs stronger-than-text or retrieval-aware signals; future work should test richer features, larger and more diverse corpora, and decision rules that exploit retrieved context before committing to a chunk size.

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A Main Results

This section collects the headline test-split numbers summarised in §3.4 and §4.1. Table 1 reports the fixed-size baselines and the per-question oracle, and Table 2 adds the RRF, type-aware, and learned-router systems together with their oracle-gap closures. All systems are evaluated on the same $n = 84$ test split.

Table 1: Fixed-size baselines and per-question oracle on the test split ($n = 84$, action space $\{128, 256, 512\}$). The oracle picks the best chunk size per question by F1 (tie-break: EM, then smaller size). The oracle–best-baseline gap is +8.19 F1 points.

System	n	F1	EM	Faithfulness
Fixed (size = 128)	84	0.2128	0.1429	0.3626
Fixed (size = 256)	84	0.1701	0.0714	0.3635
Fixed (size = 512)	84	0.1695	0.0833	0.3140
Oracle	84	0.2947	–	–

Table 2: Router evaluation on the test split ($n = 84$, action space $\{128, 256, 512\}$). The router is selected by a two-pass procedure: top-3 grid cells by val macro-F1 are re-ranked by val RAG F1. Gap closure = (router F1 – best baseline F1) / (oracle F1 – best baseline F1).

System	F1	EM	Clf macro-F1	Gap closure
Fixed (size = 128)	0.2128	0.1429	–	0.00
Oracle	0.2947	–	–	1.00
Type-aware heuristic	0.2291	–	–	+0.20
Router (minilm + logreg)	0.1710	0.1071	0.2917	–0.51

Figure 1 places each system on the accuracy–cost plane (mean retrieved tokens per question against mean F1). RRF fusion is the only system above the Fixed (128) line, and only by paying multi-index retrieval cost; the learned router sits at single-index cost but below the baseline, and the two larger fixed sizes fall furthest from the oracle ceiling.

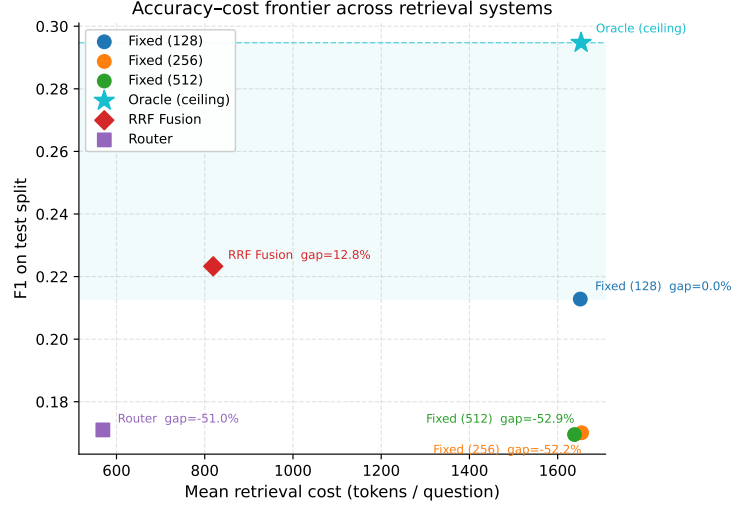


Figure 1: Accuracy-cost frontier across retrieval systems on the test split: mean retrieval cost (tokens per question) against mean F1, with each system’s oracle-gap closure annotated. RRF fusion is the only system to exceed the Fixed (128) baseline, and only at higher retrieval cost, while the learned router matches single-index cost yet falls below the best fixed baseline.

Figure 2 shows the same mean-F1 values as bars against the oracle ceiling of 0.2947, making the ordering explicit: oracle > type-aware (0.2291) > Fixed (128) (0.2128) > router (0.1710) $\approx$ Fixed (256/512). Only the oracle and the type-aware heuristic clear the Fixed (128) bar.

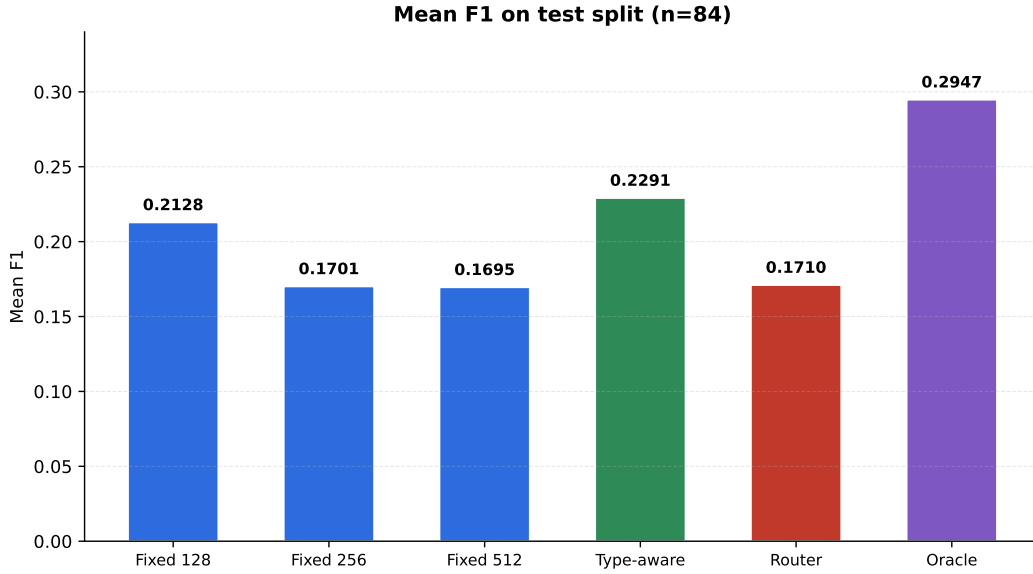


Figure 2: Mean F1 on the test split for all evaluated systems, relative to the oracle ceiling.

B Oracle Headroom and Class Imbalance

The router’s difficulty traces to two distributional facts: where the oracle-best labels concentrate, and where the routable headroom actually lives.

Figure 3 reports the share of questions for which each size is oracle-best, by split. Size 128 dominates every split (83.1%/88.3%/82.1% on train/validation/test), so any classifier faces a heavily imbalanced target and a strong majority-class baseline that predicting non-128 must beat.

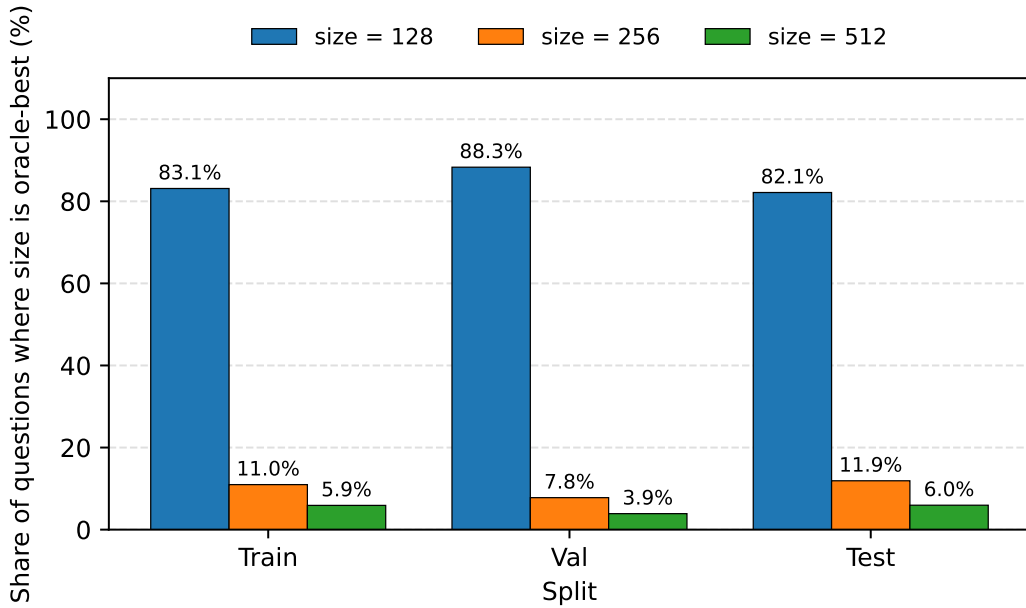


Figure 3: Share of questions for which each chunk size is oracle-best, by split. Size 128 dominates every split (82.1% on test), illustrating the class imbalance the router must overcome.

Figure 4 breaks per-question oracle F1 down by type. Factoid questions have a high mean oracle F1 (0.63) with a bimodal spread, whereas multihop (0.16) and synthesis (0.10) are low and tightly concentrated near zero: almost all recoverable headroom sits in factoids, the type a small chunk already serves well.

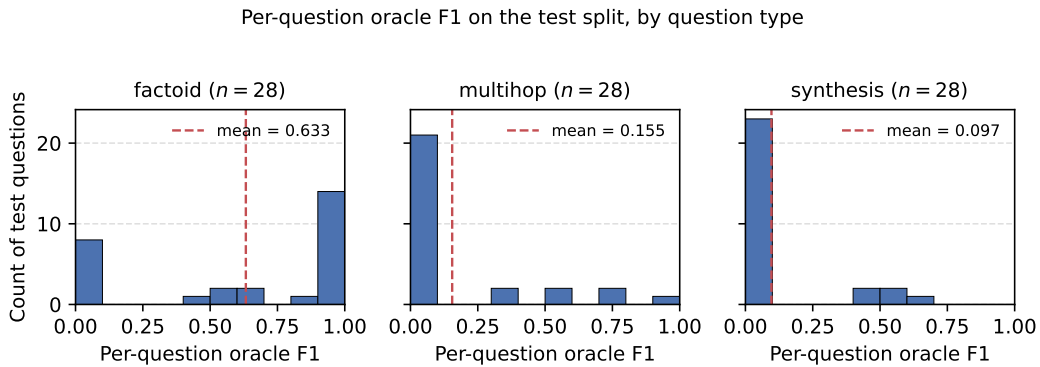


Figure 4: Distribution of per-question oracle F1 on the test split, by question type; factoid questions carry most of the routable headroom.

Figure 5 compares per-type mean F1 for the best fixed baseline, the type-aware heuristic, and the trained router. Within each type the three are nearly indistinguishable, confirming that once question type is fixed the router neither helps nor hurts appreciably.



Figure 5: Per-question-type mean F1 on the test split for the best fixed baseline, the type-aware heuristic, and the trained router (MiniLM + LR).

C Action-Space Restriction

Excluding size 1024 from the router target space is justified on two grounds. Economically, 1024 is the weakest and most expensive single-size system on the test split: its F1 is 0.1573, below 128 (0.2128), 256 (0.1701), and 512 (0.1695), while its mean context cost is highest (Table 4). Statistically, 1024 is a low-signal class — in the full four-size grid it is the oracle winner for only 6.0% of test questions — so adding it would only deepen the label imbalance with an action that is dominated on average. We therefore use the four-size grid (Table 3) only to document the cost of excluding 1024.

Table 3: Action-space ablation on the test split. The full action space includes the retired chunk size 1024; the restricted action space is what the router actually uses. Restricting to three sizes sacrifices -2.42 overall F1 points of oracle headroom, concentrated in multihop and synthesis. We retain the 1024 chunks, FAISS index, and eval-grid rows so this comparison remains exactly reproducible.

Type	Full (4 sizes)	Restricted (3 sizes)	Sacrifice
overall	+10.61	+8.19	-2.42
factoid	+15.82	+14.63	-1.19
multihop	+5.82	+3.02	-2.80
synthesis	+5.32	+2.05	-3.27

Table 4: Extended per-size fixed-baseline metrics on the test split, all four chunk sizes (including the retired 1024). Cost is the mean context-token count per question reported by the metrics module.

Chunk size	n	F1	EM	Faithfulness	Mean cost (tokens)
128	84	0.2128	0.1429	0.3626	1651
256	84	0.1701	0.0714	0.3635	1654
512	84	0.1695	0.0833	0.3140	1638
1024 (retired)	84	0.1573	0.0714	0.3092	1962

Figure 6 plots the full-versus-restricted oracle gaps. Dropping 1024 costs -2.42 overall gap points, concentrated in multihop (-2.80) and synthesis (-3.27) while factoid loses only -1.19 : the sacrificed headroom falls precisely in the types where the router already struggles, which is why the restriction is cheap in practice.

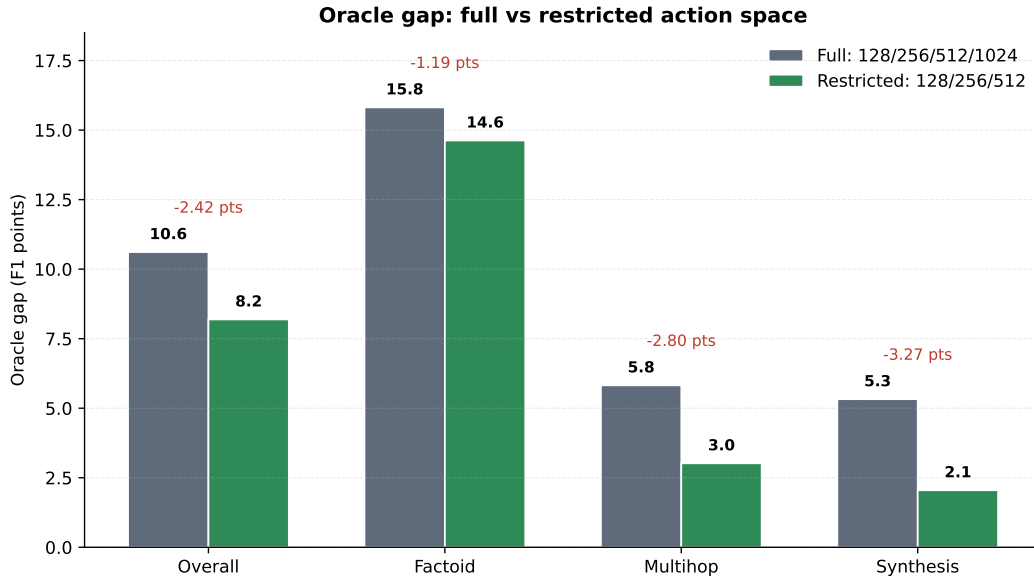


Figure 6: Oracle gap (F1 points) for the full four-size action space versus the restricted three-size space, overall and by question type.

D Natural-Prior Router Follow-up

The router reported in the body (§3.3, Tables 1 and 2) trains with inverse-frequency class weights and decodes by argmax, yielding test gap closure of -51.0% . We ran a follow-up configuration that reverses both choices: natural class priors (no reweighting) plus a confidence-threshold override at inference. The override predicts the argmax over non-majority classes only when $P(\text{non-majority}) > \tau$, else falls back to the majority class 128. The threshold τ is selected on the validation split by maximising validation RAG F1, with ties broken toward larger τ .

Threshold selection. The validation sweep over $\tau \in \{\text{None}, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}$ returned $\tau = 0.7$ as the chosen operating point. All eight sweep values tied at validation RAG F1 = 0: at no threshold did the classifier exceed the confidence bar for a non-majority class on any validation question, so every prediction defaulted to size 128 and the per-question RAG outputs were identical to Fixed (128) at every τ . The tie-break rule therefore selected the largest threshold.

Test-set behaviour. On the test split, the same pattern holds: the router predicts size 128 for all 84 test questions, fraction overridden = 0.000, and downstream RAG F1, EM, faithfulness, and mean retrieval cost coincide with Fixed (128) by construction. Gap closure is exactly 0.000. Classifier-level metrics (accuracy =

0.821, balanced accuracy = 0.333, classification macro-F1 = 0.301) confirm the collapse: accuracy equals the majority-class prior, and balanced accuracy equals chance for a three-class problem.

Figure 7 shows the resulting collapse onto the Fixed (128) point, and Tables 5 and 6 give the operating-point comparison and the full validation sweep.

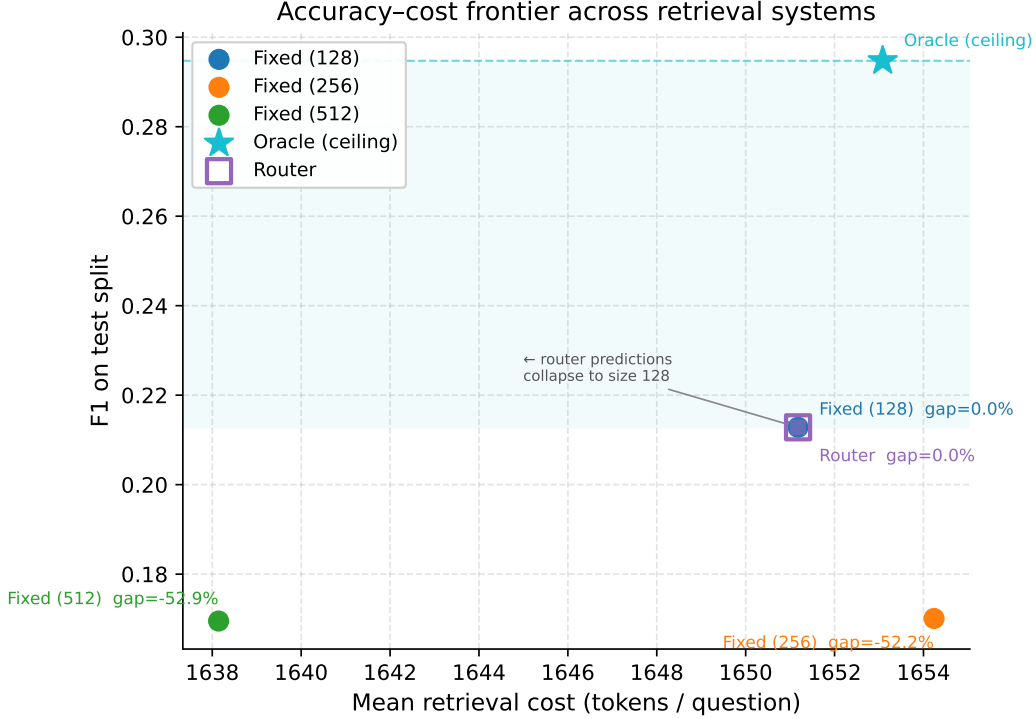


Figure 7: Accuracy-cost frontier for the natural-prior router ($\tau = 0.7$): its predictions collapse entirely onto chunk size 128, so it coincides with the Fixed (128) baseline (gap closure 0.0%).

Interpretation. The natural-prior router exhibits the opposite failure mode from the balanced-prior router in the body: where balanced weights drove the router *below* Fixed (128) by predicting minority classes incorrectly (-51.0% gap closure), natural priors collapse the router *onto* Fixed (128) by never predicting minority classes (0.0% gap closure). Neither configuration recovers oracle headroom, and no intermediate τ in the validation sweep produced a non-trivial operating point. Combined with the type-aware heuristic’s $+20\%$ gap closure in Table 2, this strengthens the body’s interpretation: the residual routable signal in this dataset is captured at the question-type level, and the trained router does not extract additional signal from query text under either weighting regime.

Operating point	F1	Non-128 (%)	Clf. acc.	Clf. bal. acc.	Gap closure
Balanced + argmax	0.1710	–	0.595	0.299	-0.510
Natural prior + $\tau = 0.7$	0.2128	0.0	0.821	0.333	0.000

Table 5: Router operating-point ablation on the test split ($n = 84$). The two configurations share the same feature pipeline, oracle labels, and validation/test splits; they differ only in (a) class-weight handling at training time and (b) the inference-time confidence threshold τ . Neither operating point exceeds the Fixed (128) baseline at F1 = 0.2128. The “Non-128” column is the fraction of test questions for which the router predicted a chunk size other than 128; it is unreported for the balanced configuration as it was not logged at that run.

Threshold τ	Val RAG F1	Fraction non-majority
None (argmax)	0.000	0.000
0.40	0.000	0.000
0.45	0.000	0.000
0.50	0.000	0.000
0.55	0.000	0.000
0.60	0.000	0.000
0.65	0.000	0.000
0.70	0.000	0.000

Table 6: Validation threshold sweep for the natural-prior router. At every τ , the classifier never produced a non-majority prediction on any validation question, so validation RAG F1 is identical to Fixed (128) at every operating point. The tie-break rule selects the largest τ .

E QA Benchmark Construction

The 398 validated QA pairs are produced by a synthetic-generation-then-filter pipeline; all settings below are fixed in `configs/base.yaml`. Questions are generated from size-512 source chunks with a one-chunk neighbour window and target an even split across the three types. Question generation uses `gpt-5.4-mini` for factoid questions and `gpt-5.5` for multihop and synthesis questions; answer generation and the validation judge use `gpt-5.5`. Sampling temperatures are 0.4 for questions, 0.1 for answers, and 0.0 for the judge.

Generated pairs pass a two-stage filter. Stage 1 discards any pair whose answer the answer-generator cannot reproduce even when given the source chunk, using a primary-only token-F1 threshold of 0.8. Stage 2 is an LLM judge that issues a KEEP/DROP decision per pair, with up to three top-up rounds that regenerate to maintain the per-type quota. Of the candidate pool, 99 pairs were rejected and 398 retained (133 factoid, 133 synthesis, 132 multihop). All downstream scoring (EM, token-F1, faithfulness) is deterministic and uses no LLM judge, so the evaluation numbers in the body do not depend on any generative model at test time. The full prompt templates are provided in the project repository under `prompts/`. Table 7 lists the generation and filtering hyperparameters, and Table 8 gives the validated set’s composition by split and type.

Setting	Value
Question models	<code>gpt-5.4-mini</code> (factoid), <code>gpt-5.5</code> (multihop, synthesis)
Answer / judge model	<code>gpt-5.5</code>
Question / answer / judge temperature	0.4 / 0.1 / 0.0
Source chunk size	512
Neighbour window	1
Target type distribution	1/3 factoid / multihop / synthesis
Stage-1 primary-only F1 threshold	0.8
Max top-up rounds	3
Rejected / validated pairs	99 / 398

Table 7: QA-generation and filtering configuration (`configs/base.yaml`).

Table 8: Validated QA-set composition by split and question type. Stratification is by question type with seed 42. Total is 398 (not 400) due to a now-fixed offset bug in the QA-id allocator that produced two duplicate ids; the multihop duplicates were dropped to keep the eval grid content-consistent.

Split	factoid	multihop	synthesis	total
train	80	78	79	237
val	25	26	26	77
test	28	28	28	84
total	133	132	133	398

Table 9: Question-Answer Composition

F Classifier Selection and Calibration

The classifier ablation compares balanced logistic regression, linear SVM, and LightGBM under the feature families of §3.3. Logistic regression is selected, consistent with the small, imbalanced training set (237 examples, oracle labels skewed toward size 128), which leaves higher-capacity decision boundaries little minority-class evidence to exploit. Probabilities are calibrated on the SVM path via Platt scaling when class counts permit it (Platt 1999, pp. 61–74), falling back to a softmax over decision scores otherwise. Because the headline metric is the hard chunk-size decision, calibrated probabilities cannot by themselves recover the oracle gap.

G Reproducibility

All headline numbers in this paper are deterministic under the configuration in Table 11: rerunning `compute_oracle` followed by `run_baselines` regenerates the eval grid and the headline tables from the same seed, embedding model, generator, and scoring scripts.

Table 10: Reproducibility setup. All numbers in this paper are deterministic under this configuration; running `python -m experiments.compute_oracle` followed by `python -m experiments.run_baselines` regenerates the eval grid and the headline tables.

Setting	Value
Random seed	42
Config file	<code>configs/base.yaml</code> (local), <code>configs/cluster.yaml</code> (A100)
Corpus	<code>data/corpus.txt</code>
Embedding model	BAAI/bge-small-en-v1.5
Embedding device (cluster)	cuda (A100 MIG slice)
Generation backend (cluster)	vLLM 0.6.3.post1
Generator	Qwen/Qwen2.5-7B-Instruct (bf16, max_model_len = 8192)
Generation temperature	0.0
Max new tokens	256
Action-space sizes	128, 256, 512
Retrieval k per size	128: 20, 256: 10, 512: 5
Splits ratios	0.6/0.2/0.2 (stratified by type)
QA validated count	398
QA validated SHA-256	c062b3c31f8ec80c...

Table 11: Reproducibility

Dataset Construction Sources

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